Semantic-Guided Imitation Learning

***Abstract*—**

***Index Terms*—Imitation learning, semantic representations, visuomotor control.**

I. Introduction

Reliable execution of sequential manipulation tasks is essential for practical robotic applications but remains challenging [1]. Sequential manipulation tasks require a robot to achieve an overall goal through a sequence of interrelated operations [2]. Such tasks arise in practical applications ranging from household cleaning, where a tablecloth is lifted to clear debris beneath it and then restored, to surgery, where tissue is retracted to expose the target region before resection [3]. A key challenge of these tasks is that preceding interactions change the conditions under which subsequent operations can be performed. For example, compared with simple pick-and-place tasks involving visible and accessible targets, manipulation under occlusion requires the robot to assess whether prior actions have made the target visible and accessible or further manipulation is needed [4], [5]. Consequently, reliable execution requires a policy that adapts subsequent actions to interaction outcomes observed through visual feedback.

Visuomotor imitation learning offers a practical approach to sequential manipulation by reducing the need for explicit interaction modeling, but reliable execution remains difficult with limited demonstrations. It learns a policy that maps visual and proprioceptive observations to actions using expert demonstrations. For example, Action Chunking with Transformers (ACT) [6] has been used in Mobile ALOHA to learn coordinated mobile manipulation, including opening a cabinet and storing a cooking pot [7]. Similarly, Diffusion Policy [8] has been adopted in the Universal Manipulation Interface (UMI) framework to learn bimanual cloth folding and multi-step dish washing [9]. However, execution errors can lead a policy into states poorly covered by limited demonstrations, where its predicted actions may no longer suit the actual task conditions, compromising subsequent operations and task completion [10].

Prior work has focused on improving the robustness of visuomotor policies to execution deviations by expanding data coverage, improving visual representations, and adapting actions during execution. Ross et al. [10] expanded the training dataset with expert action labels for states visited by the learned policy, while Laskey et al. [11] injected noise into expert actions during demonstration collection to obtain examples of recovery behavior. Although data coverage expansion improves coverage of states arising from execution deviations, it requires additional expert supervision or modifications to demonstration collection, limiting its applicability when only a fixed demonstration set is available. Zhu et al. introduced object-centric visual representations for robust visuomotor imitation [12] and incorporated 3D object representations to improve generalization across backgrounds, viewpoints, and object instances [13]. Although these methods improve generalization to visual variations, robust object representations alone do not ensure that subsequent actions remain appropriate when execution deviates from demonstrations.

In this work, we develop a theory-guided visuomotor imitation learning framework that uses task-relevant representation learning to support sequential manipulation from limited demonstrations. Our analysis derives conditions linking imitation error and representation quality to task-failure bounds. Guided by this analysis, we supervise policy query tokens with task descriptors derived from semantic masks, encouraging the policy to retain information relevant to changing manipulation conditions. The policy is trained from the fixed demonstration set, without additional robot interaction for policy training. Simulation and real-world experiments demonstrate the effectiveness of the proposed framework in sequential manipulation tasks.

II. Related Work

III. Problem Formulation and Theoretical Analysis

This section studies conditions for improving task performance from limited demonstrations. Starting from the relationship between imitation error and task success, we examine how to preserve information needed to distinguish the task consequences of different actions.

We first establish conditions under which action error bounds task-failure probability and explain why improved coverage in representation space alone does not guarantee better task performance. We then examine how semantic guidance can preserve task-relevant information and, under consistent expert selection, control imitation-error transfer through expert-action reconstruction.

A. Visuomotor Imitation Learning

We study offline visuomotor imitation learning from a finite, fixed set of robot demonstrations, for manipulation tasks whose completion can be assessed from visual history. The policy generates control commands from visual and proprioceptive observations and available execution history. Training is based on this fixed demonstration dataset, with demonstrated actions providing the primary imitation target, whereas deployment performance is measured by the probability of task completion over a finite horizon.

1) System and Observations: Consider a robotic system governed by a single decision policy that performs a given manipulation task through the joint actions of its actuators. The joint robot-environment state $s_t \in \mathcal{S}$ evolves under the command $u_t \in \mathcal{A}$ and gives rise to observation o_t. The observation $o_t = (I_t^{1:V}, q_t)$ comprises multi-view images

$I_t^{1:V} = (I_t^1, \ldots, I_t^V)$ and proprioceptive information q_t, where $V \geq 1$ is the number of cameras and I_t^v is the image from the vth view. Over a finite horizon of T control steps, this interaction is represented by the following partially observed controlled process:

$$\begin{aligned} s_0 &\sim p_0, \\ s_{t+1} &\sim P(\cdot \mid s_t, u_t), \quad 0 \leq t < T, \\ o_t &\sim O(\cdot \mid s_t), \quad 0 \leq t \leq T. \end{aligned} \tag{1}$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A} \subseteq \mathbb{R}^{d_u}$ is the admissible command set specified by the chosen robot control interface, d_u is the command dimension, p_0 is the initial-state distribution, and P and O are the transition and observation kernels, respectively.

2) Offline Demonstrations and Policy: The policy learns from a fixed demonstration set $\mathcal{D} = \{\zeta_i\}_{i=1}^N$ for the same manipulation task. The ith demonstration trajectory ζ_i records observations and demonstrated control commands:

$$\zeta_i = (o_0^i, u_0^{\mathrm{E},i}, o_1^i, \ldots, u_{T_i-1}^{\mathrm{E},i}, o_{T_i}^i). \tag{2}$$

Here, N is the finite number of demonstrations, $T_i \geq 1$ is the number of control steps in the ith trajectory, and the superscript E denotes demonstrated commands.

The information available to the deployed policy π is represented by the history $\mathcal{H}_t$. This history includes observations $o_{0:t}$ obtained up to time t, previously executed commands $u_{0:t-1}$, and other runtime records before the current decision. The policy defines a distribution over admissible control commands, with $u_t \sim \pi(\cdot \mid \mathcal{H}_t)$. The command history is empty at $t = 0$.

3) Learning and Task Objectives: For the following analysis, we consider deterministic policies acting on the available history, with π_{E} denoting the expert policy. Let ν denote the population distribution underlying the demonstrations, where $(t, h) \sim \nu$ specifies a control step and its available history. We define the population imitation risk as the expected squared Euclidean deviation from the expert's action:

$$R_\nu(\pi) := \mathbb{E}_{(t,h)\sim\nu} \|\pi(h) - \pi_{\mathrm{E}}(h)\|^2. \tag{3}$$

Training seeks to reduce this risk through an empirical imitation loss on the finite demonstrations; the specific training losses are defined in Section IV.

The deployment objective is to complete the task within the finite horizon. The measurable task-completion function $\Phi : \mathcal{S}^{T+1} \to \{0, 1\}$ equals 1 for state trajectories satisfying the task's terminal or temporal requirements and 0 otherwise. This defines the completion indicator Ψ and the task-completion probability $J(\pi)$ of the deployed policy π:

$$\Psi = \Phi(s_{0:T}), \qquad J(\pi) = \mathbb{P}_\pi(\Psi = 1). \tag{4}$$

Here, $\mathbb{P}_\pi$ is the trajectory probability law induced jointly by the system model and the complete deployed policy π. Learning aims to reduce $R_\nu(\pi)$, whereas deployment aims to increase $J(\pi)$. A lower imitation risk does not by itself guarantee a higher task-completion probability; the following analysis examines conditions relating these objectives.

4) Visual Assessability: Under a fixed camera configuration and evaluation horizon T, the evaluation domain $\Omega_{\text{eval}} \subseteq \mathcal{S}^{T+1} \times \mathcal{I}^{T+1}$ consists of prespecified pairs of physical trajectories and visual histories consistent with the system model and visibility conditions, where $\mathcal{I}$ is the space of multi-view image tuples $I_t^{1:V}$.

Assumption 1 (Visual assessability): There exists a measurable visual decision function $\mathcal{V} : \mathcal{I}^{T+1} \to \{0, 1\}$ such that, for every $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$,

$$\Phi(s_{0:T}) = \mathcal{V}(I_{0:T}^{1:V}). \tag{5}$$

Assumption 1 specifies visual identifiability of completion within the evaluation domain, without requiring full-state observability. The subsequent analysis considers policy executions and counterfactual action comparisons for which $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$ almost surely.

B. From Imitation Error to Task Performance

Whether action error bounds task failure depends on where deviations occur and their subsequent task consequences. We establish this connection and show that transferring imitation risk from demonstrations to deployment depends jointly on representation-level coverage and conditional action-error discrepancy.

1) Task-Performance Difference Between Policies: We compare a policy π with a reference policy π_0 through their task-failure probabilities and express the difference as a sum of expected single-step action consequences. Each consequence is defined by comparing alternative actions at the same history while following π_0 afterward [14], [15]. This decomposition provides the basis for relating imitation error to task performance.

We consider deterministic policies acting on the available history. Given a history $\mathcal{H}_t = h$ and a candidate action $a \in \mathcal{A}$, define the action-value function as the probability of task failure ($\Psi = 0$):

$$Q_t^{\pi_0}(h, a) := \mathbb{P}^{a,\pi_0}(\Psi = 0 \mid \mathcal{H}_t = h)\,. \tag{6}$$

Here, $\mathbb{P}^{a,\pi_0}$ denotes the probability law obtained by taking action a at time t and following π_0 afterward.

When the current action is also selected by π_0, this reduces to the value function of π_0, $V_t^{\pi_0}(h) := Q_t^{\pi_0}\big(h, \pi_0(h)\big)$. The effect of choosing a instead of $\pi_0(h)$ is therefore measured by

$$A_t^{\pi_0}(h, a) := Q_t^{\pi_0}(h, a) - V_t^{\pi_0}(h). \tag{7}$$

Thus, $A_t^{\pi_0}(h, a) > 0$ indicates an increase in failure probability, whereas $A_t^{\pi_0}(h, a) < 0$ indicates a decrease.

To relate these action-level consequences to differences in task performance between policies, we require the following common comparison conditions.

Assumption 2 (Common task and environment): The compared policies share the same initial distribution and task-completion criterion. Given a history h and an action a, the next available history follows a conditional distribution that does not otherwise depend on the policy. The reference policy π_0 is defined on all histories involved in the comparison.

Let $F(\pi) = 1 - J(\pi)$ denote the task-failure probability of policy π. To evaluate π relative to π_0, we set $a = \pi(h)$ and aggregate the resulting action consequences along trajectories generated by π.

Applying the finite-horizon performance-difference identity [14]–[16] to these one-step action effects gives

$$F(\pi) - F(\pi_0) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) \right]. \tag{8}$$

Here, $\mathbb{E}_\pi$ denotes expectation over trajectories generated by π. Equation (8) evaluates action effects at the situations actually visited by the evaluated policy π. Its proof is given in Appendix A-A. This relation allows task-performance differences to be bounded through the consequences of action deviations.

2) Task-Failure Bounds from Imitation Risk: Taking the expert policy as the reference, we show that the imitation risk $R_\nu(\pi)$ bounds task-failure probability $F(\pi)$ when demonstrations sufficiently cover deployment histories and action deviations have bounded task consequences.

Imitation risk is evaluated under the demonstration distribution ν, but deployment histories may follow a different distribution because actions depend on π. Let μ_π^t denote the distribution of $(t, \mathcal{H}_t)$ at control step t under π. The distribution $\bar{\mu}_\pi = T^{-1} \sum_{t=0}^{T-1} \mu_\pi^t$ corresponds to choosing a control step uniformly and observing its history.

Assumption 3 (Demonstration coverage):

For the evaluated policy π, there exists a uniform constant $1 \le C < \infty$ such that

$$\bar{\mu}_\pi(\mathcal{B}) \le C\, \nu(\mathcal{B})$$

for every measurable set $\mathcal{B}$ of pairs (t, h).

This condition excludes deployment situations absent from the reference distribution and limits how much their probabilities can be underestimated.

Assumption 4 (Action sensitivity): Let A_t^{E} denote the action effect with the expert policy as the reference. For a fixed action scale, there exists a uniform constant $0 \le L < \infty$ such that

$$|A_t^{\mathrm{E}}(h, a)| \le L \|a - \pi_{\mathrm{E}}(h)\|$$

for all relevant (t, h, a).

This assumption allows action deviations to be translated into bounds on changes in task-failure probability.

Following the connection between deployment imitation error and task cost [10], Assumptions 3 and 4, together with the performance-difference identity, yield the following bound (Appendix A-B):

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C R_\nu(\pi)}. \tag{9}$$

Here, C accounts for the shift from demonstration to deployment distributions, while L measures the sensitivity of task outcomes to action deviations. With other quantities fixed, a smaller $R_\nu(\pi)$ yields a tighter upper bound on task-failure probability.

This bound identifies coverage and action sensitivity as key conditions for translating imitation accuracy into task-performance guarantees. We next examine the limitations of complete-history coverage and the role of representations in risk transfer.

3) Imitation-Risk Transfer Through Representations: To address the limitations of complete-history coverage, we analyze imitation-risk transfer in representation space. We derive a bound showing that compression can relax coverage requirements, while differences in conditional imitation error must also be accounted for.

Assumption 3 can fail even after an arbitrarily small action deviation. For $T \ge 2$, suppose the expert always executes $u_0 = 0$, whereas the learned policy executes an admissible $u_0 = \varepsilon \ne 0$. Histories recording $u_0 = \varepsilon$ then occur under the learned policy but have zero probability under the expert demonstration distribution. No finite coverage constant C can satisfy the assumption.

A representation can merge such distinctions, potentially relaxing coverage requirements, but may also discard information needed to distinguish action consequences. This tension between distributional alignment and information loss under a noninvertible representation also appears in domain-adaptation analyses [17], [18].

For a fixed policy, expert, and common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index, let μ_Z and ν_Z be the representation distributions induced by $\bar{\mu}_\pi$ and ν. In place of Assumption 3, suppose $\mu_Z(\mathcal{B}) \le C_\phi \nu_Z(\mathcal{B})$ for every measurable representation event $\mathcal{B}$, with $1 \le C_\phi < \infty$. Even at the same representation, the two distributions may contain different histories. Let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$ and define their conditional-error discrepancy as $D_\phi = \mathbb{E}_{z \sim \mu_Z} |m_\mu(z) - m_\nu(z)|$, where $m_\mu(z) = \mathbb{E}_{\bar{\mu}_\pi}[e(h)^2 \mid Z = z]$ and $m_\nu(z) = \mathbb{E}_\nu[e(h)^2 \mid Z = z]$. With square-integrable action error under both distributions and measurable versions of these conditional means,

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C_\phi R_\nu(\pi) + D_\phi. \tag{10}$$

If Assumption 4 also holds, the performance-difference identity gives

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + D_\phi}. \tag{11}$$

Appendix A-C provides the proof. With imitation risk and action sensitivity fixed, (11) shows that the effect of a representation on the bound depends jointly on the coverage factor C_ϕ and the conditional error discrepancy D_ϕ. This motivates seeking representations that reduce C_ϕ while controlling D_ϕ, so that their combined contribution $C_\phi R_\nu(\pi) + D_\phi$ decreases.

C. Task-Relevant Information Preservation

We seek conditions under which a representation can improve coverage while controlling the conditional imitation-error discrepancy. We examine semantic guidance as a way to retain task-relevant information, then connect the accuracy of that information to expert-action reconstruction and a bound on D_ϕ.

1) Semantic-Guided Compression and Coverage: Semantic guidance can encourage an encoder to merge appearance differences while retaining object-level structure relevant to action selection. This motivates a representation design, not an automatic coverage guarantee. As in Section III-B3, compression can relax coverage requirements by making some history

distinctions irrelevant to representation events. Appending a deterministic semantic map while retaining the original RGB input does not itself change the best coverage constant relative to RGB, since projection recovers that input. The relevant object is the learned decision representation $Z = \phi(t, \mathcal{H}_t)$, not the number of input modalities.

Improved coverage alone does not ensure a tighter bound; the error associated with merging histories must also be accounted for. A sufficient route is to retain information from which a common rule can reconstruct expert actions under both demonstration and deployment distributions. This emphasis on decision-relevant information is consistent with information-state and behaviorally grounded representation analyses [19], [20]. We next formulate this requirement using object attributes and spatial relationships.

2) Task Information and Expert-Action Reconstruction: We examine when retained task information is sufficient to approximate expert choices. The reconstruction error depends on both what the selected information fails to explain and how accurately it is estimated.

Fix the evaluated policy π and set $\mu = \bar{\mu}_\pi$, with ν the demonstration distribution. Write $\pi(h) = p(Z)$ for the same measurable policy readout under both distributions. We define a task descriptor vector $G \in \mathbb{R}^{d_G}$ whose components describe task-relevant object attributes and spatial relationships. Semantic masks provide a way to construct such quantities, including visibility and relative position. Let χ denote task-specified proprioceptive and execution context, selected in advance rather than identified with the unrestricted history. Suppose Z retains χ and supports an estimate $\widehat{G} \in \mathbb{R}^{d_G}$ of G. For $\rho \in \{\mu, \nu\}$, define $\delta_{G,\rho} = \mathbb{E}_\rho \|G - \widehat{G}\|$. These quantities must be defined under both distributions, with an explicit convention for unavailable components.

Assumption 5 (Descriptor-conditioned expert selection): There exists a common measurable rule g_{E} mapping descriptors and context to $\mathcal{A}$ such that, for $\rho \in \{\mu, \nu\}$,

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - g_{\mathrm{E}}(G, \chi)\| \le \epsilon_{\mathrm{sel},\rho}. \tag{12}$$

For a finite constant $K_{\mathrm{E}} \ge 0$,

$$\|g_{\mathrm{E}}(G, \chi) - g_{\mathrm{E}}(\widehat{G}, \chi)\| \le K_{\mathrm{E}} \|G - \widehat{G}\| \tag{13}$$

holds almost surely under both distributions, with finite $\epsilon_{\mathrm{sel},\rho}$ and $\delta_{G,\rho}$.

Here, $\epsilon_{\mathrm{sel},\rho}$ measures expert choices unexplained by (G, χ), while K_{E} controls sensitivity to estimation error. Task relevance alone does not imply this condition: equally successful actions may reflect different expert choices.

Define $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality yields

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - \widehat{a}_{\mathrm{E}}(Z)\| \le b_\rho := \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}} \delta_{G,\rho}. \tag{14}$$

Appendix B-A gives the derivation. Supervising descriptor readouts targets the estimation term, while the choice of descriptors and context determines how well expert actions can be explained.

3) Bounds on Conditional Imitation-Error Discrepancy: Expert-action reconstruction connects information preservation to the discrepancy in Section III-B3. Because both $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z)$ depend on the same representation, their squared difference is a common function under μ and ν. Expert reconstruction error bounds how far the actual conditional imitation errors depart from this function.

Assume a finite action-set diameter $B_{\mathcal{A}} = \sup_{a,a' \in \mathcal{A}} \|a - a'\|$ under a fixed action scale.

Proposition 1 (Conditional-error transfer through expert reconstruction): Under Assumption 5 and the representation-level coverage condition in Section III-B3, with b_ρ defined in (14),

$$D_\phi \le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu). \tag{15}$$

If Assumption 4 also holds,

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu)}. \tag{16}$$

The proof is in Appendix B-B. The design objective is therefore to improve coverage while preserving information needed to reconstruct expert actions. For a fixed expert baseline, horizon, and action sensitivity, reducing the combined imitation and reconstruction terms tightens the task-failure bound.

IV. Method

Section III motivates preserving information relevant to expert action selection. Figure 1 organizes this approach into two components. First, quality-aware iterative annotation expands a small manually reviewed seed set into semantic supervision. Second, RGB observations, semantic maps, and supervised task-descriptor queries jointly support a base action-chunk policy.

A. Quality-Aware Iterative Semantic Supervision

Task-descriptor labels are derived from semantic masks, so segmentation errors can compromise the supervision used to preserve task information. We combine reviewed seed annotations with video-level quality screening to limit error propagation as the labeled set expands (Fig. 2).

1) Event-Guided Seeds and Iterative Expansion: We select a small set of videos spanning the principal interaction and occlusion states, and annotate keyframes around events such as target exposure, contact, and placement. Bounding boxes and positive and negative point prompts guide bidirectional SAM2 propagation [21]; consistency checks, class-dependent structural processing, and manual correction produce the seed masks. The seeds train an initial U-Net segmenter, which predicts candidate masks on new unlabeled videos. Accepted masks accumulate across iterations, and each subsequent segmenter is reinitialized and trained on this cumulative pool without the original seed videos. The seed videos then serve as a fixed reference for monitoring segmentation accuracy and temporal anomalies. Iteration continues while anomaly rates improve without a material loss of segmentation accuracy.

2) Video-Level Quality Screening: The screening procedure distinguishes sustained task changes from transient segmentation anomalies. For each class, we extract trajectories of mask area, spatial extent, centroid, and topology, and fit robust piecewise trends over the video. Persistent changes can establish new segments, while short-lived departures provide anomaly evidence instead of penalizing every inter-frame

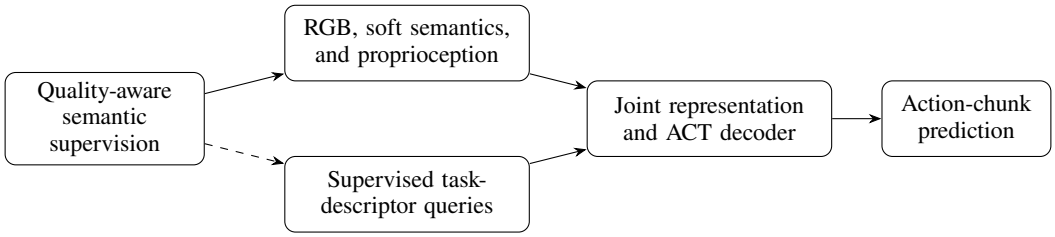


Fig. 1. Method overview. Quality-aware iterative annotation provides a frozen segmenter and quality-weighted descriptor labels. RGB, soft semantics, proprioception, and supervised queries jointly support action-chunk prediction.

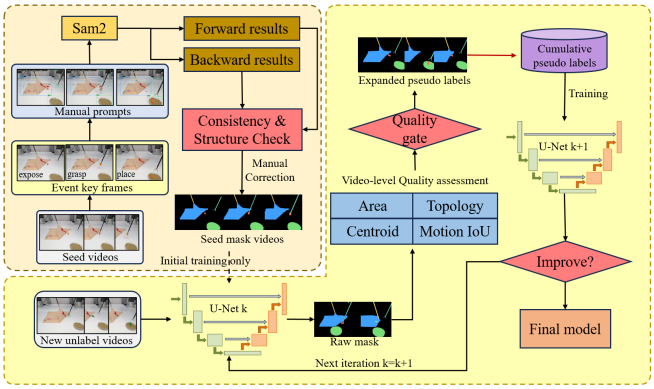


Fig. 2. Quality-aware iterative annotation. Reviewed bidirectional SAM2 masks train the initial segmenter. Subsequent models learn from cumulative, quality-screened pseudo-labels; seed videos are retained for reference rather than further training.

change. Motion-compensated mask IoU supplies additional temporal evidence on pixels with forward–backward-consistent optical-flow correspondences. For class c at frame t, we collect these diagnostics as

$$\boldsymbol{\upsilon}_{t,c}^{\text{seg}} = (\upsilon_{t,c}^{\text{area}}, \upsilon_{t,c}^{\text{space}}, \upsilon_{t,c}^{\text{topo}}, \upsilon_{t,c}^{\text{motion}}). \tag{17}$$

Agreement across evidence groups identifies anomalous intervals; excessive changes introduced by structural processing also trigger exclusion. A frame enters subsequent segmentation training only when its relevant classes pass these checks. The manually reviewed reference set complements temporal screening by checking for persistent semantic errors that temporal consistency alone cannot detect. The same diagnostics also determine descriptor-specific supervision weights in Section IV-B. Quality assessment is performed offline; the final per-view segmenters are frozen and require only current images during policy execution.

B. Semantic Inputs and Task-Descriptor Queries

Semantic maps emphasize object-level spatial structure, while supervised queries encourage the joint representation to retain task-relevant attributes and relationships. RGB and proprioception remain available for decision cues not captured by the semantic classes. This design targets the descriptor-estimation term in (14) through the learned representation, rather than assuming that adding semantic channels improves coverage.

1) Joint Visual-Semantic Encoding: For view v, a frozen segmenter predicts class probabilities $p_t^v(c,\xi)$ from image I_t^v. We form a soft semantic image as

$$S_t^v(\xi) = \sum_{c \in \mathcal{C}_v} p_t^v(c,\xi) e_c^{\text{sem}}, \tag{18}$$

where $\mathcal{C}_v$ is the view-specific class set, ξ a pixel location, and $e_c^{\text{sem}} \in [0,1]^3$ a fixed class color. This probability-weighted color representation allows RGB and semantics to use the same three-channel visual encoder without argmax hardening. The encoded streams from all views are combined with proprioceptive and query tokens in a transformer encoder. An ACT decoder [6] reads this context to predict the base action chunk.

2) Supervised Descriptor Queries: We select K_{rel} descriptor groups $G^{(j)} \in \mathbb{R}^{d_j}$, such as visible area and relative position, which together form the task descriptor vector G. Fixed geometric operators applied to the offline masks generate pseudo-labels $\widetilde{G}_t^{(j)}$; the task-specific definitions are given in Appendix C-C. Each group has a learnable query whose encoded output $z_t^{(j)}$ supports a prediction head, $\widehat{G}_t^{(j)} = D_j(z_t^{(j)})$, and remains available to the action decoder. Supervision therefore encourages task information to be recoverable from the policy representation without prescribing an attention map.

A weight $\varpi_{n,j}^G \in [0,1]$ combines descriptor validity with quality evidence from the relevant semantic classes. Reliable labels receive full weight, uncertain labels are downweighted, and undefined or anomalous labels receive zero weight. In particular, a distance is undefined when either object is invisible, whereas a confirmed absence can provide a valid zero-visible-area label. Disabling a descriptor loss does not remove the sample's action supervision. With descriptors normalized to fixed scales, the training objective is

$$\mathcal{L}_{\text{rel}} = \frac{1}{K_{\text{rel}}} \sum_j \frac{\sum_n \varpi_{n,j}^G \ell_G(\widehat{G}_n^{(j)} - \widetilde{G}_n^{(j)})}{\max(1, \sum_n \varpi_{n,j}^G)},$$
$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{IL}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}. \tag{19}$$

Here, n indexes minibatch samples, ℓ_G is the component-averaged Smooth L1 loss, and $\mathcal{L}_{\text{IL}}$ is the standard ACT training objective. The segmenters remain frozen, and quality weights affect auxiliary supervision rather than online policy inputs. Appendices C-A and C-B specify label admission and its connection to descriptor-estimation error.

V. 实机实验

实验围绕三个问题展开：完整方法相对现有动作生成策略的表现如何，语义输入与查询监督分别带来何种收益，以及这些收益如何随物体类别和初始位置变化。全部结果来自完成补采后的五组实机记录，共 480 次有效试验。

A. 任务、数据与评价协议

双臂系统执行遮挡物体操作：一侧操作遮挡布使目标暴露，另一侧使用工具将目标移出遮挡并送入指定区域，随后恢复遮挡布覆盖。输入来自前、侧两个 RGB 视角，统一为 640×360。示范数据包含 108 条轨迹、61740 帧，以 30 Hz 记录 10 维关节位置和动作，训练物体为方块、纸球和螺丝。测试增加训练未见的蜡笔，按相同任务目标评估类别变化下的表现。这里的训练未见指动作示范中的物体类别，不涉及对视觉预训练语料的排除承诺。

每类物体设置 12 个位置，每位置重复两次，每种方法共 96 次。位置以 (x, y) 表示，$x = 0, 1, 2, 3$ 从图像左向右排列，$y = 0, 1, 2$ 从上向下排列。每种方法在同一位置汇总四类物体后有 8 次试验；每类物体有 24 次。该坐标用于描述测试网格，不赋予行号物理深度含义。

单次策略控制时限为 40 s，目标控制频率为 30 Hz。统一视觉评估器使用前视角检测目标暴露、脱离遮挡、进入目标区域及遮挡布恢复，侧视角只作诊断。完成条件要求上述状态同时成立，并在最近 7 次分割评估中至少获得 4 次成功票；遮挡布恢复使用前视角像素占比阈值 0.32。最终统计采用保存并由操作者确认的成功或失败标签。阶段检测是辅助证据，不以「曾到达某阶段」替代最终成功。所有纳入记录均为正常退出且未手动中止；一次无效中止已用正常完成的补采记录替换。附录 D列出标签修正和补采处理。

B. 本轮测试的实现配置

本轮最终 checkpoint 采用过滤后的硬标签语义图，训练读取离线标签，在线使用冻结 U-Net 的 argmax 输出和连通分量过滤。三个任务查询分别对应可见性、工具–目标向量和目标–区域距离，每个查询包含前、侧视角独立读出头，共 8 个标量。前视角工具参考点为主轴左端点，侧视角为工具质心；区域距离使用目标像素到固定区域的最大归一化距离。辅助 Smooth L1 损失的尺度为 0.01，分母为几何有效坐标数，质量权重直接缩放分子。这些是本轮实验的实际配置；方法与方法附录中保留的早期表示、几何和损失形式未作为本轮实现参数使用。

C. 对照方法与消融设置

ACT 与 Diffusion Policy（DP）为基线。基于 ACT 构建四种模块配置：ACT 仅使用 RGB 与本体信息；RGBQtoken 增加关系查询监督；segNoQtoken 增加双视角语义输入；viewQtoken 同时使用语义输入与关系查询，为完整方法。图中分别简写为 ACT、RGB + Q、语义无 Q 和语义 + Q。这四种配置构成语义输入与查询监督的 2×2 对照。

四种 ACT 配置每次预测 60 步并执行 30 步，采用同步推理。DP 预测 64 步、执行 30 步，采用异步推理；其 DDIM 去噪步数在三种训练物体上为 16，在蜡笔上为 24。DP 比较体现相应部署配置的表现。所有配置均关闭额外修正模块；本组结果不包含 VIOLA。

主要指标为固定试验集合上的成功次数与成功率，报告总体 Wilson 95% 区间作为比例不确定性的描述。方法比较以物体–位置为区组，在每个区组内合并两方法各两次结果，以超几何分配卷积计算双侧成功次数差检验。总体五方法的 10 个两两比较采用 Holm 校正。试验编号不作为跨方法的一一配对依据。检验以区组内结果可交换为零假设条件，未将采集协议表述为随机分配。

D. 总体表现与物体类别

表 I和图 3显示，完整方法成功 92/96 次，达到 95.83%。ACT、DP、RGBQtoken 及 segNoQtoken 分别为 54/96、50/96、72/96 和 60/96。完整方法相对四组对照分别提高 39.58、43.75、20.83 和 33.33 个百分点。总体区组检验经 Holm 校正后，完整方法与四组对照的差异均达到 0.05 水平（表 II）。RGBQtoken 相对 ACT 和 DP 的校正后 p 值分别为 0.0188 与 0.000845；segNoQtoken 相对 ACT 的差异未达到该水平，校正后 $p = 0.817$。

完整方法在四类物体上均为 23/24，训练物体合计 69/72，蜡笔 23/24，二者观察成功率同为 95.83%。ACT 和 DP 在三种训练物体上均为 41/72，在蜡笔上分别为 13/24 和 9/24。RGBQtoken 与 segNoQtoken 在蜡笔上分别为 19/24 与 17/24，完整方法进一步提高到 23/24。按相同类别比较，完整方法相对 ACT 的增量分别为 5、11、12 和 10 次成功，相对 segNoQtoken 为 7、12、7 和 6 次，纸球上的改进最明显。对四类物体全部 40 个方法比较统一校正后，完整方法相对 segNoQtoken 在纸球上仍有差异证据（$p = 0.00376$）；不能将总体显著性直接推广为每个类别的单独显著性。

E. 语义输入与查询监督的消融

图 5给出四种 ACT 配置。相对 RGB 基线，仅增加查询监督提高 18.75 个百分点，仅增加语义输入提高 6.25 个百分点。在查询监督存在时增加语义输入，提高 20.83 个百分点；在语义输入存在时增加查询监督，提高 33.33 个百分点。完整方法相对两个单模块配置分别多成功 20 次和 32 次。

两条增益之差为 14.58 个百分点，描述了本次试验中联合配置的额外改善。本研究未对这一交互差异实施独立推断，因此将其作为描述性结果。实验支持联合设计优于两个单模块配置，不能仅凭该差值断言存在显著的统计交互效应。

F. 位置分布与同条件重复表现

图 4按相同位置汇总四种物体，附录图 8进一步给出每个物体–位置的两次结果。完整方法 12 个位置均至少成功 7/8 次；ACT 范围为 1/8 至 8/8，DP 为 0/8 至 8/8，RGBQtoken 为 3/8 至 8/8，segNoQtoken 为 1/8 至 8/8。因此，完整方法的观察优势体现在更多位置上保持较高成功次数。

按列汇总，每列 24 次，完整方法从左到右为 23、24、23、22 次成功，ACT 为 6、17、18、13，DP 为 2、9、19、20，RGBQtoken 为 12、18、20、22，segNoQtoken 为 8、17、18、17。完整方法相对 RGBQtoken 新增的 20 次成功中，17 次来自左侧两列。按行汇总，每行 32 次，完整方法从上到下为 30、30、32，segNoQtoken 为 16、14、30。二者 32 次成功数增量中有 30 次来自上方两行；segNoQtoken

TABLE I
五组方法的最终实机结果。单元格为成功次数/试验数（成功率，%）；蜡笔为训练未见物体。

方法	方块	纸球	螺丝	蜡笔	总体
ACT	18/24 (75.00)	12/24 (50.00)	11/24 (45.83)	13/24 (54.17)	54/96 (56.25)
DP	12/24 (50.00)	16/24 (66.67)	13/24 (54.17)	9/24 (37.50)	50/96 (52.08)
RGBQtoken	18/24 (75.00)	16/24 (66.67)	19/24 (79.17)	19/24 (79.17)	72/96 (75.00)
segNoQtoken	16/24 (66.67)	11/24 (45.83)	16/24 (66.67)	17/24 (70.83)	60/96 (62.50)
viewQtoken	**23/24 (95.83)**	**23/24 (95.83)**	**23/24 (95.83)**	**23/24 (95.83)**	**92/96 (95.83)**

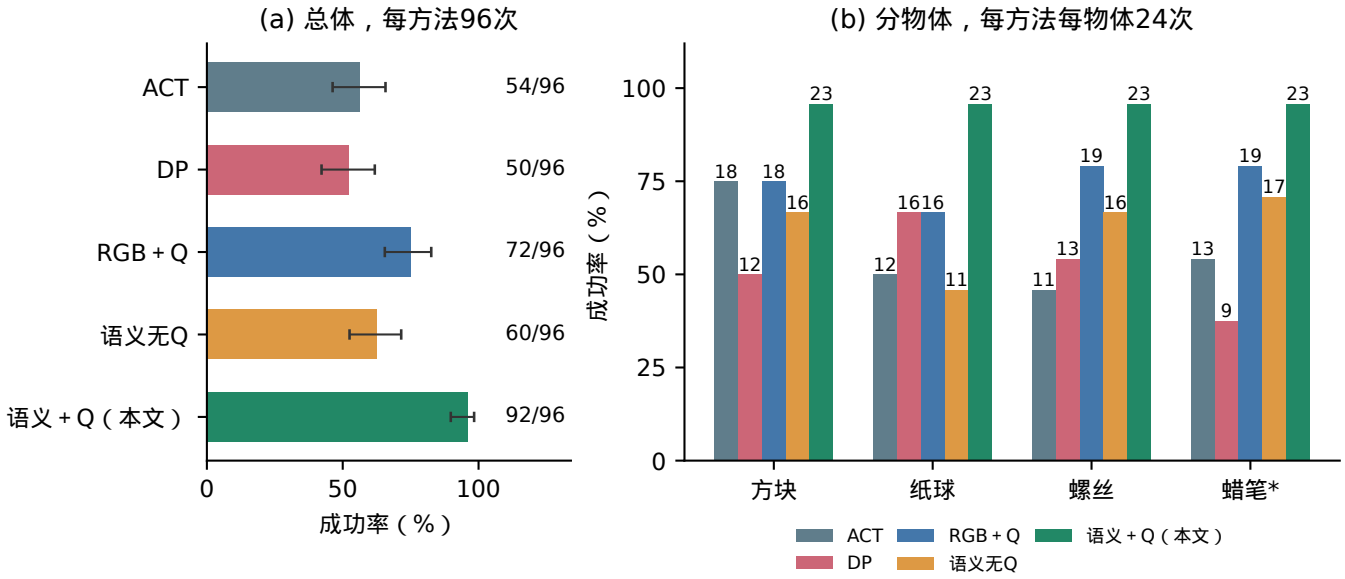


Fig. 3. 总体及分物体比较。(a) 每方法 96 次，误差线为总体比例的 Wilson 95% 区间；右侧标注成功次数。(b) 每方法每类物体 24 次，柱顶数字为成功次数；蜡笔标记为训练未见类别。完整方法在四类物体上均为 23/24。

TABLE II
完整方法相对各对照的总体差异。成功率差以百分点计；p 值在全部 10 个总体比较内校正。

对照	成功率差	Holm 校正 p
ACT	39.58	3.83×10^{-10}
DP	43.75	8.57×10^{-12}
RGBQtoken	20.83	1.15×10^{-4}
segNoQtoken	33.33	2.21×10^{-8}

的 36 次失败中有 34 次位于这两行。这些分布说明，加入查询监督后的改善并非仅来自原本容易的位置。

对位置、物体、列和行分别进行分层置换关联检验，使用 200000 次置换，对五方法共 20 项检验统一 Holm 校正。DP 与 segNoQtoken 的 12 位置关联分别得到 $p = 0.000100$ 和 $p = 0.00144$；ACT、DP、RGBQtoken 及 segNoQtoken 的列关联校正后 p 分别为 0.0217、0.000100、0.0480 和 0.0258。DP 与 segNoQtoken 还表现出行关联，校正后 p 分别为 0.0480 和 0.000270。完整方法的四类关联检验均未检出差异；五种方法的物体类别关联均未达到校正后 0.05 水平。未检出关联不等于证明位置或类别无影响，尤其不能仅凭不同方法各自的 p 值判定方法间的位置敏感性差异显著。

48 个物体–位置条件中，两次均成功的条件数依次为 ACT 17、DP 18、RGBQtoken 31、segNoQtoken 22 和完整方法 44。完整方法无两次均失败的条件，其余 4 个条件均为一成一败，分别是方块 $(3,1)$、纸球 $(3,0)$、螺丝 $(2,0)$ 和蜡笔 $(0,1)$。在相同条件的成功次数比较中，完整方法相对 segNoQtoken 在 24 个条件较好、23 个相同、1 个较差；相对 RGBQtoken 为 16 个较好、30 个相同、2 个较差。因此，整体改善并不意味着逐条件严格占优。

G. 完成时间与失败阶段

表 III及图 6同时报告时间和失败分布。完整方法 20 s 内成功 86/96 次，30 s 内成功 92/96 次；RGBQtoken 相应为 69/96 与 72/96，segNoQtoken 为 55/96 与 57/96。成功试验的平均控制时长分别为完整方法 14.77 s、RGBQtoken 13.55 s、segNoQtoken 13.80 s、ACT 14.96 s 与 DP 15.28 s。各方法成功子集不同，完整方法并非成功子集平均耗时最短者。将全部试验纳入后，完整方法平均控制时长为 15.82 s，反映其更少进入超时失败。

按「曾暴露、曾分离、曾入区、最终保存成功」顺序，完整方法为 96、96、92、92 次，segNoQtoken 为 94、93、62、60，RGBQtoken 为 93、92、73、72。ACT 和 DP 分别为 90、89、55、54 及 93、93、50、50。在 segNoQtoken 的 36 次失败中，31 次到达分离阶段但未入区；完整方法的 4 次失败均属于这一阶段。因此，完整方法的主要改善出现在分离后的目标入区过程，而剩余失败也集中于此。「曾入区

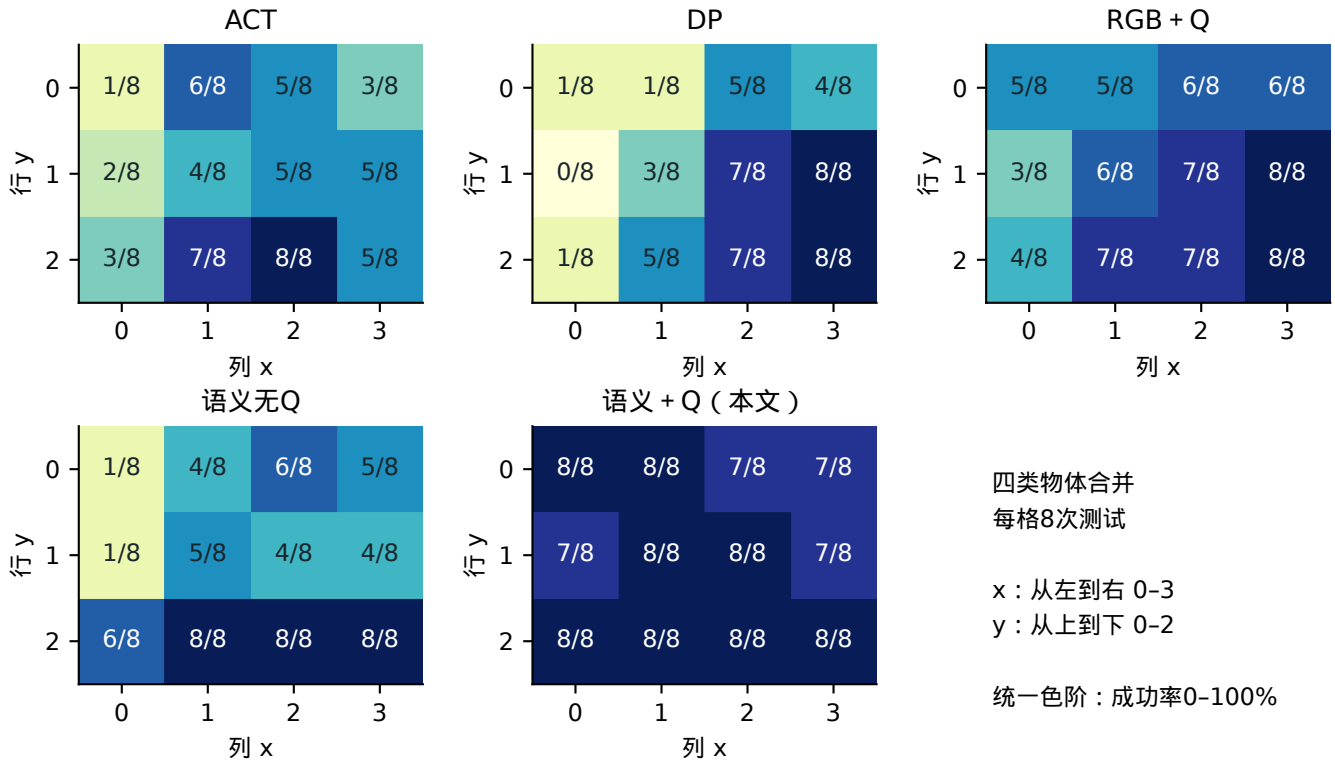


Fig. 4. 相同位置下的跨物体汇总。每格为成功次数/8，统一色阶对应 0–100% 成功率；横轴从左向右、纵轴从上向下。位置 $(0,1)$ 上，ACT、DP、RGBQtoken 和 segNoQtoken 分别为 2/8、0/8、3/8 和 1/8，完整方法为 7/8。

TABLE III
总体比例区间与控制时长。时间不含标签保存及视频编码；成功子集平均值与全部试验平均值分别统计。

方法	成功率 95% 区间（%）	成功子集均时（s）	全部试验均时（s）	20 s 内成功
ACT	46.28–65.74	14.96	25.92	48/96
DP	42.20–61.80	15.28	27.13	40/96
RGBQtoken	65.49–82.59	13.55	20.17	69/96
segNoQtoken	52.51–71.53	13.80	23.43	55/96
viewQtoken	89.77–98.37	14.77	15.82	86/96

但最终失败」包含后续条件未满足以及人工标签修正，不能统一解释为恢复遮挡失败。

日志中完整方法的可训练参数为 51592082 个，相比 ACT 的 51584906 个增加 7176 个，约 0.014%。冻结分割器使总参数量增加至 76531328 个。完整方法加权平均动作块推理流水线耗时约 79.24 ms，ACT 约 12.16 ms，DP 约 255.48 ms，RGBQtoken 约 54.15 ms，segNoQtoken 约 71.20 ms。这些日志值包含各自流水线开销，不能作为仅由查询头造成的延迟或统一硬件基准；共享执行前缀和目标频率也不等于各方法实际控制节拍完全一致。

VI. Discussion and Conclusion

A. Failure Cases and Limitations

本节结合表 IV的失败分布和同步视频、指令、感知日志，分析完整方法尚未解决的问题。案例选择覆盖不同失败类型，不以个别成功片段替代整体统计。

1) Perception and Task Information: 质量筛选针对视频中的异常变化，不能排除时间上稳定的系统性误分。所选描述量刻画对象属性与空间关系，但不一定足以解释接触状态或专家的具体动作选择；语义分割准确与描述量读出准确也不意味着策略已经正确利用这些信息。

（结合图 7(a,b) 区分分割错误、描述量估计错误和执行决策错误。说明是否存在描述量准确仍失败的情况，并联系理论中的专家选择近似误差解释未被当前信息覆盖的动作选择因素。没有探针或干预证据时，将原因表述为待验证解释，而非确定归因。）

2) Scope of the Evidence: 本文的任务表现界使用总体模仿误差，并依赖覆盖、动作敏感性和专家选择条件。独立轨迹上的读出和动作误差用于检验相应机制，不能将有限训练损失直接代入这些界，也不能据更紧的上界确定实际成功率排序。两个任务上的分别训练验证方法的适用性，不等同于零样本跨任务迁移；手术模拟材料上的实验亦不构成临床有效性证据。

（根据真实划分与训练重复说明结论覆盖哪些物体、位置和平台。联系理论 B–C 总结哪些机制获得实验支持、哪些条件尚未得到直接测量；概括新增任务的描述量设计、语义标注与训练成本。若未做多预算实验，不声明样本效率优势；QToken 没有独立停止头时，不将任务成功等同于自主停止能力。）

TABLE IV
按最远到达阶段划分的失败计数。各列互斥，合计等于保存为失败的试验数。

方法	未暴露	暴露未分离	分离未入区	曾入区但最终失败	总失败
ACT	6	1	34	1	42
DP	3	0	43	0	46
RGBQtoken	3	1	19	1	24
segNoQtoken	2	1	31	2	36
viewQtoken	0	0	4	0	4

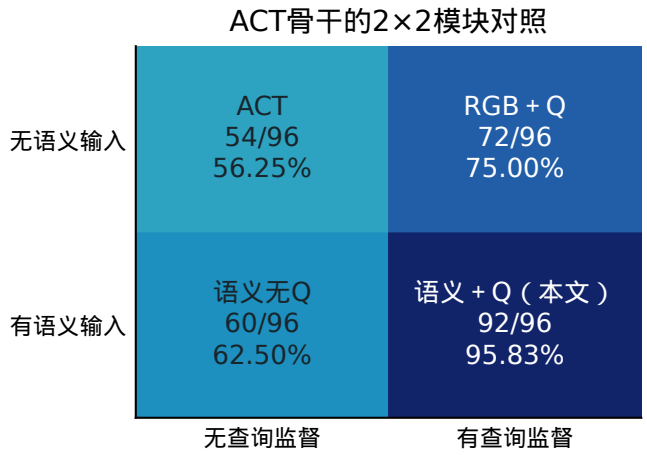


Fig. 5. 语义输入与查询监督的四格对照，每格 96 次。所有配置保留 RGB 和本体观测，语义输入为额外视觉流。

B. Conclusion

（用一段总结问题、两个方法模块及两个实机任务支持的核心结论。填写实际成功率、稳定性或恢复收益，只保留与匹配对照相符的表述；最后指出一个由失败分析直接支持的主要局限，不预先声明显著提升、通用性或收敛保证。）

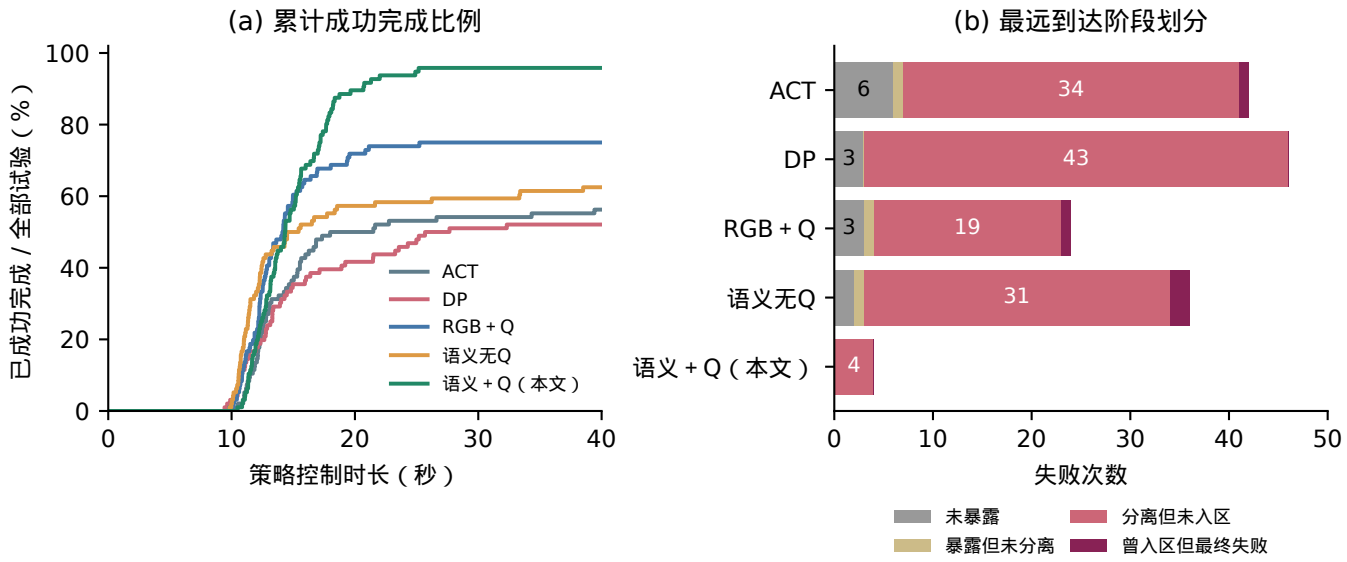


Fig. 6. 完成时间与失败阶段。(a) 分母固定为每方法全部 96 次试验，失败试验不会从分母移除。(b) 依据记录中最远到达的自动阶段与最终保存标签划分失败；边界误判经人工改为失败，保留在最后一类中。

(a) 持续分割错误：RGB、语义预测、人工审核
(b) 描述量准确但执行失败：任务关系与接触状态
填写两个任务的代表性案例；只标注日志与视频支持的事实

Fig. 7. Representative failure cases and supporting evidence. Placeholder for perception and task-information limitations.

Appendix A

A. *Proof of the Policy Performance-Difference Identity*

We prove (8) by summing expected changes in the reference value. The evaluation-domain restriction in Section III-A and Assumption 1 give the terminal value

$$V_T^{\pi_0}(\mathcal{H}_T) = 1 - \mathcal{V}(I_{0:T}^{1:V}) = 1 - \Psi. \tag{20}$$

For $t < T$, taking action $\pi(\mathcal{H}_t)$ and then following π_0 gives the failure probability $V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})$ conditional on the next history. Averaging over next histories under Assumption 2 gives

$$Q_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = \mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]. \tag{21}$$

Here, π determines the next-history distribution, while $V_{t+1}^{\pi_0}$ accounts for subsequent execution under π_0.

Combining (7) and (21) and taking expectations gives

$$\begin{aligned}&\mathbb{E}_\pi[A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))] \\ &= \mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)].\end{aligned} \tag{22a}$$

By the tower property, averaging the conditional next-step value over current histories gives its unconditional expectation:

$$\begin{aligned}&\mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right] \\ &= \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})].\end{aligned} \tag{22b}$$

Substituting (22b) into (22a) and summing over time leaves only the terminal and initial terms:

$$\begin{aligned}&\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) \\ &= \sum_{t=0}^{T-1} \left\{\mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)]\right\} \\ &= \mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] - \mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)].\end{aligned} \tag{22c}$$

At time T, no action remains. By (20), the terminal term is the failure probability under π:

$$\mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] = \mathbb{E}_\pi[1 - \Psi] = F(\pi). \tag{22d}$$

The policies share the initial-history distribution, so the initial term equals the failure probability of following π_0 throughout:

$$\begin{aligned}\mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)] &= \mathbb{E}_{\pi_0}[V_0^{\pi_0}(\mathcal{H}_0)] \\ &= F(\pi_0).\end{aligned} \tag{22e}$$

Finally, substituting (22d) and (22e) into (22c) gives

$$\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = F(\pi) - F(\pi_0). \tag{22f}$$

This establishes (8).

B. Proof of the Task-Failure Bound

We prove (9) by first bounding task consequences through action deviations and then transferring the resulting deployment error to the demonstration distribution.

Set $\pi_0 = \pi_{\mathrm{E}}$ and let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$. Applying Assumption 4 in (8) bounds each step's increase in failure probability by its action deviation:

$$F(\pi) - F(\pi_{\mathrm{E}}) \le L\,\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right]. \tag{23a}$$

Since $\bar{\mu}_\pi$ samples control steps uniformly, the time sum equals T times the deployment average:

$$\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right] = T\,\mathbb{E}_{(t,h)\sim\bar{\mu}_\pi} e(h). \tag{23b}$$

By the Cauchy–Schwarz inequality, the mean action deviation is bounded by its root mean square:

$$\mathbb{E}_{\bar{\mu}_\pi} e(h) \le \sqrt{\mathbb{E}_{\bar{\mu}_\pi} e(h)^2}. \tag{23c}$$

Assumption 3 also bounds expectations of nonnegative measurable functions, by simple-function approximation and monotone convergence. Applying it to $e(h)^2$ and using (3) gives

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C\,\mathbb{E}_{\nu} e(h)^2 = C R_\nu(\pi). \tag{23d}$$

Combining (23a)–(23d) establishes (9).

C. Proof of the Representation-Level Transfer Bound

We prove (10) by separating conditional error discrepancy from distributional coverage, then apply action sensitivity to obtain (11).

Let $\mu = \bar{\mu}_\pi$, with the remaining notation and the integrability and measurable conditional-mean conditions as in Section III-B3. Representation coverage ensures that ν_Z-null sets are also μ_Z-null sets, so the demonstration-side conditional mean is well defined for the deployment comparison.

First, conditioning on the representation gives

$$\mathbb{E}_{\mu} e(h)^2 = \mathbb{E}_{\mu_Z} m_\mu(Z). \tag{24a}$$

At the same representation, deployment and demonstration conditional errors can differ. By the definition of D_ϕ,

$$\begin{aligned}\mathbb{E}_{\mu_Z} m_\mu(Z) &\le \mathbb{E}_{\mu_Z} m_\nu(Z) + \mathbb{E}_{\mu_Z}|m_\mu(Z) - m_\nu(Z)| \\ &= \mathbb{E}_{\mu_Z} m_\nu(Z) + D_\phi.\end{aligned} \tag{24b}$$

Finally, applying representation coverage to the nonnegative function m_ν and taking the total expectation give

$$\mathbb{E}_{\mu_Z} m_\nu(Z) \le C_\phi \mathbb{E}_{\nu_Z} m_\nu(Z) = C_\phi R_\nu(\pi). \tag{24c}$$

Combining (24a)–(24c) establishes (10). Under Assumption 4, substituting this result into the sensitivity, time-average, and Cauchy–Schwarz steps (23a)–(23c) yields (11).

Appendix B
Proofs for Task-Relevant Information Preservation

A. Proof of the Expert-Action Reconstruction Bound

We prove (14) under the common encoder and readouts of Section III-C, with $\mu = \bar{\mu}_\pi$. Define the common represented expert rule $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality and Assumption 5 give, for $\rho \in \{\mu, \nu\}$,

$$\begin{aligned}\mathbb{E}_\rho\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| &\le \mathbb{E}_\rho\|\pi_{\mathrm{E}} - g_{\mathrm{E}}(G,\chi)\| \\ &\quad + \mathbb{E}_\rho\|g_{\mathrm{E}}(G,\chi) - g_{\mathrm{E}}(\widehat{G},\chi)\| \\ &\le \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}}\delta_{G,\rho} = b_\rho.\end{aligned} \tag{25}$$

B. Proof of the Conditional-Error Transfer Bound

We prove Proposition 1 by comparing actual imitation error with a common function of Z, then transferring the demonstration-side reconstruction error through representation coverage. Use $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$ from Section III-C. Since $\pi(h) = p(\phi(t,h))$, the represented squared error $\ell_Z(z) = \|p(z) - \widehat{a}_{\mathrm{E}}(z)\|^2$ is the same function of z under both distributions. All three actions belong to $\mathcal{A}$, so the reverse triangle inequality yields

$$\begin{aligned}|e(h)^2 - \ell_Z(Z)| &\le \big(\|\pi - \pi_{\mathrm{E}}\| + \|\pi - \widehat{a}_{\mathrm{E}}\|\big)\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \\ &\le 2B_{\mathcal{A}}\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\|.\end{aligned} \tag{26}$$

Conditioning separately under each distribution gives

$$|m_\rho(z) - \ell_Z(z)| \le 2B_{\mathcal{A}}\mathbb{E}_\rho[\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \mid Z = z]. \tag{27}$$

Insert the common function ℓ_Z between m_μ and m_ν and apply representation coverage to the nonnegative demonstration-side conditional error:

$$\begin{aligned}D_\phi &\le \mathbb{E}_{\mu_Z}|m_\mu(Z) - \ell_Z(Z)| \\ &\quad + \mathbb{E}_{\mu_Z}|m_\nu(Z) - \ell_Z(Z)| \\ &\le \mathbb{E}_{\mu_Z}|m_\mu(Z) - \ell_Z(Z)| \\ &\quad + C_\phi\mathbb{E}_{\nu_Z}|m_\nu(Z) - \ell_Z(Z)| \\ &\le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu).\end{aligned} \tag{28}$$

Representation coverage ensures that the demonstration-side conditional inequality also holds μ_Z-almost everywhere. This proves (15); substitution into (11) establishes (16).

Appendix C
Method Details

A. Descriptor Labels and Quality Weights

Descriptor supervision uses the quality evidence relevant to its geometric definition rather than a single frame-wide score. Visible area depends primarily on target extent and topology; relative position also depends on the localization of both objects, and a tool-endpoint vector is sensitive to tool shape. For the classes $\mathcal{C}_j$ used by descriptor j, let $\eta^{\mathrm{seg}}_{t,c}$ indicate that class c passes anomaly screening, and let $m^G_{t,j}$ indicate that the descriptor is defined. The supervision weight is

$$\varpi^G_{t,j} = m^G_{t,j} \prod_{c\in\mathcal{C}_j} \eta^{\mathrm{seg}}_{t,c}\, g_j(v^G_{t,j}), \tag{29}$$

where $v^G_{t,j}$ aggregates the relevant evidence in (17), and g_j is a nonincreasing weight between zero and one. Reviewed

descriptor labels determine the reliable, downweighted, and rejected ranges; these weights are not probabilities of correctness. Without sufficient evidence to calibrate a graded weight, $g_j \equiv 1$ gives validity and anomaly admission alone. The groupwise normalization in (19) primarily changes the relative contributions of samples within a descriptor group.

An empty mask is a valid zero-area label only when absence is confirmed by review or supporting temporal evidence. Undefined distances and directions are excluded from the auxiliary loss rather than labeled as zero. Missing optical-flow correspondences provide no motion evidence, whereas excessive structural repair excludes a frame from segmentation training. Whole-video diagnostics may use later frames offline; they are not supplied to the deployed policy.

B. Pseudo-Label and Descriptor-Readout Errors

For either evaluation distribution ρ used in Section III-C, inserting the descriptor pseudo-label $\widetilde{G}$ gives

$$\delta_{G,\rho} \le \mathbb{E}_\rho \|\widehat{G} - \widetilde{G}\| + \mathbb{E}_\rho \|\widetilde{G} - G\|. \tag{30}$$

The terms separate descriptor fitting from pseudo-label error, motivating quality screening in addition to query supervision. When measured only on admitted labels, this comparison concerns the admitted subset; the admission rate is reported alongside the error.

C. Task-Descriptor Instance

Our task instance uses front-view masks for three descriptor groups, irrespective of whether the policy receives one or two camera views. They describe target visibility, target–region distance, and the vector from a tool endpoint to the target. For image width W and height H_I, let M_O, M_R, M_T be the object, region, and tool masks, with centroids c_O, c_R, c_T when nonempty. The labels are

$$\begin{aligned} \widetilde{G}^{(1)} &= |M_O|/(WH_I), \\ \widetilde{G}^{(2)} &= \|c_O - c_R\| / \sqrt{(W-1)^2 + (H_I-1)^2}, \\ \widetilde{G}^{(3)} &= \mathrm{clip}(\bar{c}_O - \bar{e}_{T,L}, -1, 1). \end{aligned} \tag{31}$$

Here, $\bar{c} = (c_x/(W-1), c_y/(H_I-1))$, with rightward and downward positive directions. The endpoint $\bar{e}_{T,L}$ is the leftmost of the two extremes along the tool mask's principal axis in normalized coordinates. This geometric convention need not identify the physical contact tip. Target–region centroid distance is an auxiliary descriptor, not the criterion for complete target containment.

D. Evaluation Scope

Seed videos used by the initial segmenter provide a development reference, not an independent final test. Cross-video evaluation requires held-out videos excluded from annotation-model fitting and selection, with the split maintained through policy training. Descriptor evaluation uses the base policy in deployment mode. For ACT, this sets the latent input to zero rather than conditioning on the demonstrated action posterior [6].

（定稿前须完成统一数据划分及新版描述量质量权重的实现核对；已有结果不能自动作为新版方法的验证。待核对事项单列于 method_implementation_notes.md。）

Appendix D
评估审计与分层结果

A. 有效试验与标签修正

最终数据由五种方法各 96 次有效记录构成，覆盖四类物体、12 个位置和每条件两次重复。segNoQtoken 在方块 $(1,0)$ 的一次测试中，自动评估器判为成功，操作者确认目标停在区域边缘并保存为失败；本文按保存标签统计。该组在纸球 $(3,0)$ 的一次 18.7 s 手动中止被确认为无效，已排除，并由一条正常执行至 40 s、结果为失败的补采记录替换。补采后没有缺失或重复的物体–位置–重复编号组合，全部 480 条记录均正常退出且没有手动中止标记。原有 ACT、DP、RGBQtoken 和 viewQtoken 四组记录保持不变。

B. 统计口径

跨方法比较在物体–位置区组内进行。每个区组的两方法共四个二值结果，在固定总成功数下按每方法两次重新分配；将 48 个区组的超几何分布卷积，得到总体成功次数差的精确零假设分布。分物体分析只使用对应的 12 个区组，四类物体全部 40 项方法比较构成统一校正族。

关联检验以各分组汇总成功次数的平方和为离散度统计量。位置检验在物体内置换结果，物体检验在位置内置换，列检验在物体–行内置换，行检验在物体–列内置换。使用 200000 次 Monte Carlo 置换及加一修正，最小可报告未校正 p 值约为 5×10^{-6}，不将其写为零。总体方法差异与关联检验分别使用独立的 10 项和 20 项 Holm 校正族。比例区间为总体二值计数的 Wilson 描述性区间，并未建模跨训练种子波动或逐条件的不同成功概率。

C. 完整物体–位置矩阵

图 8呈现全部 240 个方法–物体–位置单元，每格两次。同一列比较固定物体在不同位置的表现，同一行跨列比较同一方法的不同物体；相同网格坐标则用于比较不同物体在同一位置的结果。

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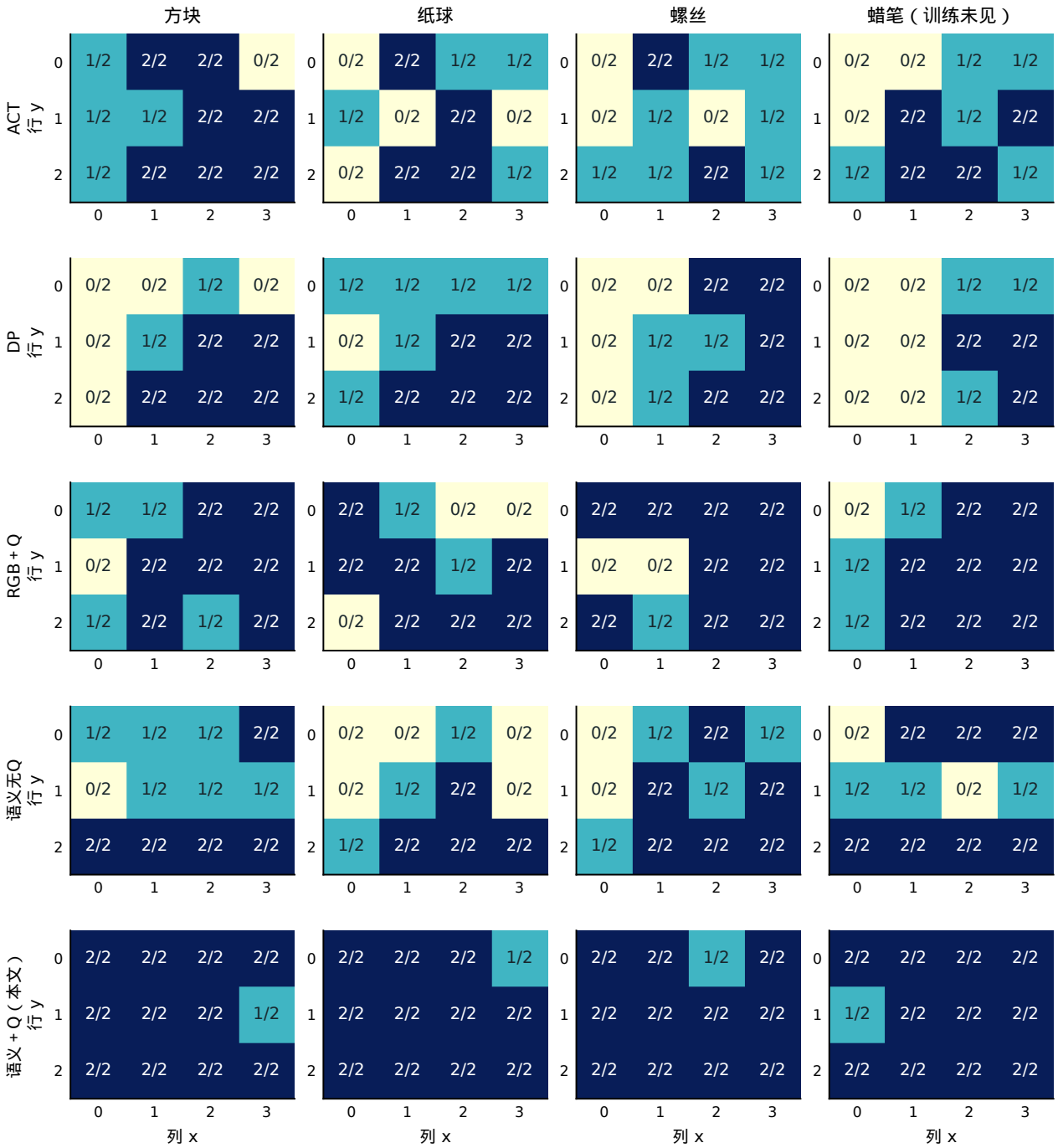


Fig. 8. 完整的物体–位置结果。每个小面板为四列三行测试网格，每格标注成功次数/2，所有面板统一色阶：浅色 0/2，中间色 1/2，深色 2/2。行对应方法，列对应物体；蜡笔为训练未见类别。

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